ICS2311 COMPUTER GRAPHIC

CAT ONE

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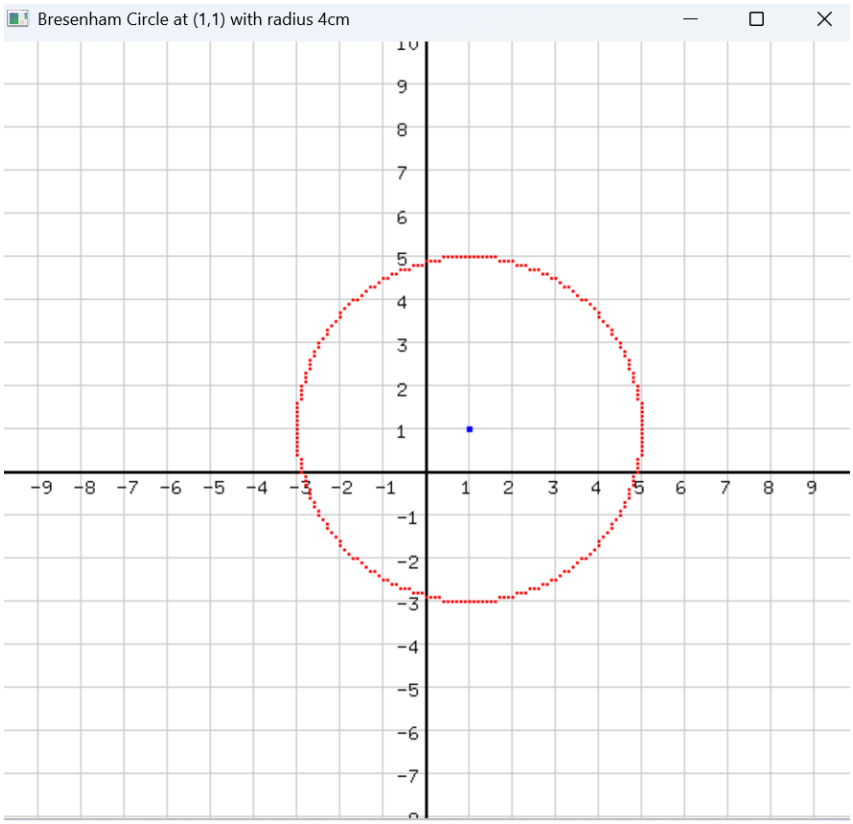
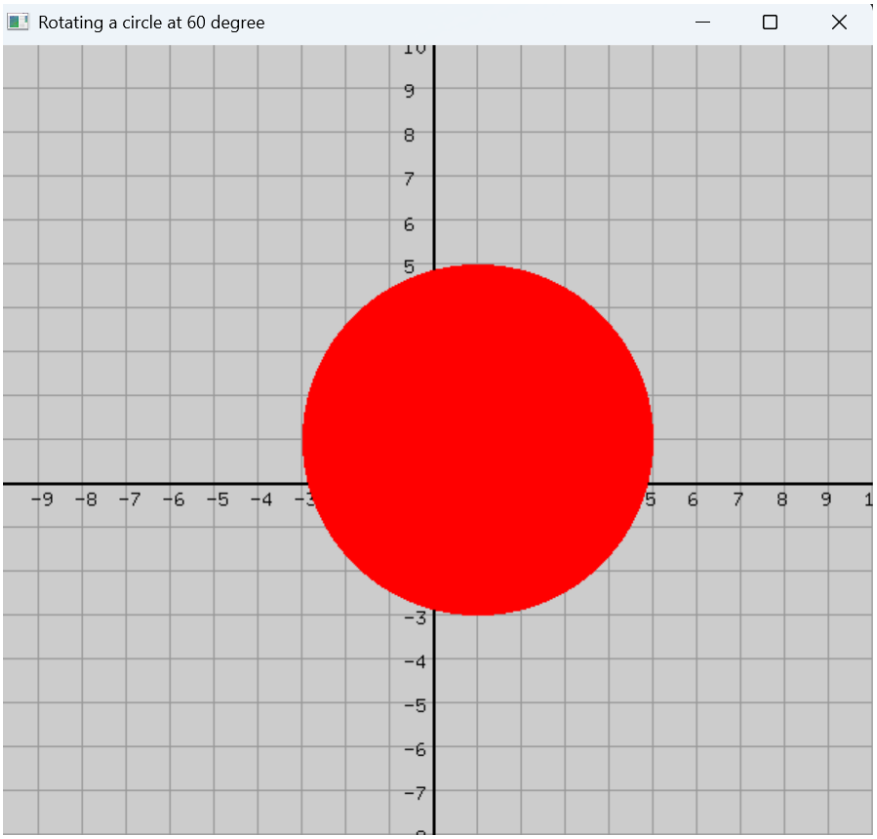
SCT211-0504/2021

QUESTION 1- Circle Drawing and triangle

a) Using OpenGL, write a program that draws a circle using Brensenham circle drawing algorithm: The parameters for the circle are as follows – the radius should be 4 Centimeters and the starting point coordinates are (1,1):

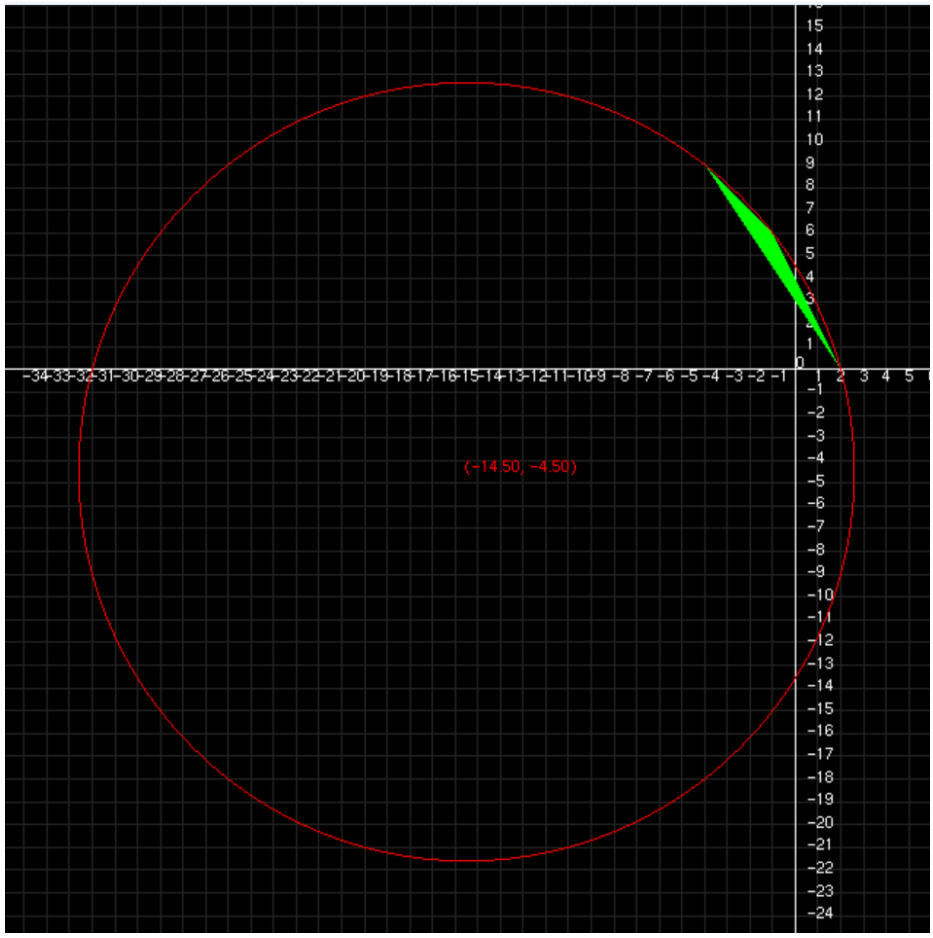


(i) write an OpenGL program to rotate the Circle 60 degrees clockwise and fill it with red shade (hint RGB # hex ff0000)



b). Using OpenGL, Write a program to draw a triangle with vertices (1,6 ; 2,0; -4,9)

i) Draw a circle round the triangle (Circumscribed)-the circle should be in red color



QUESTION 2- Ellipse drawing

a) Using Open GL and midpoint ellipse algorithm, draw an ellipse with centre as (1,2) given

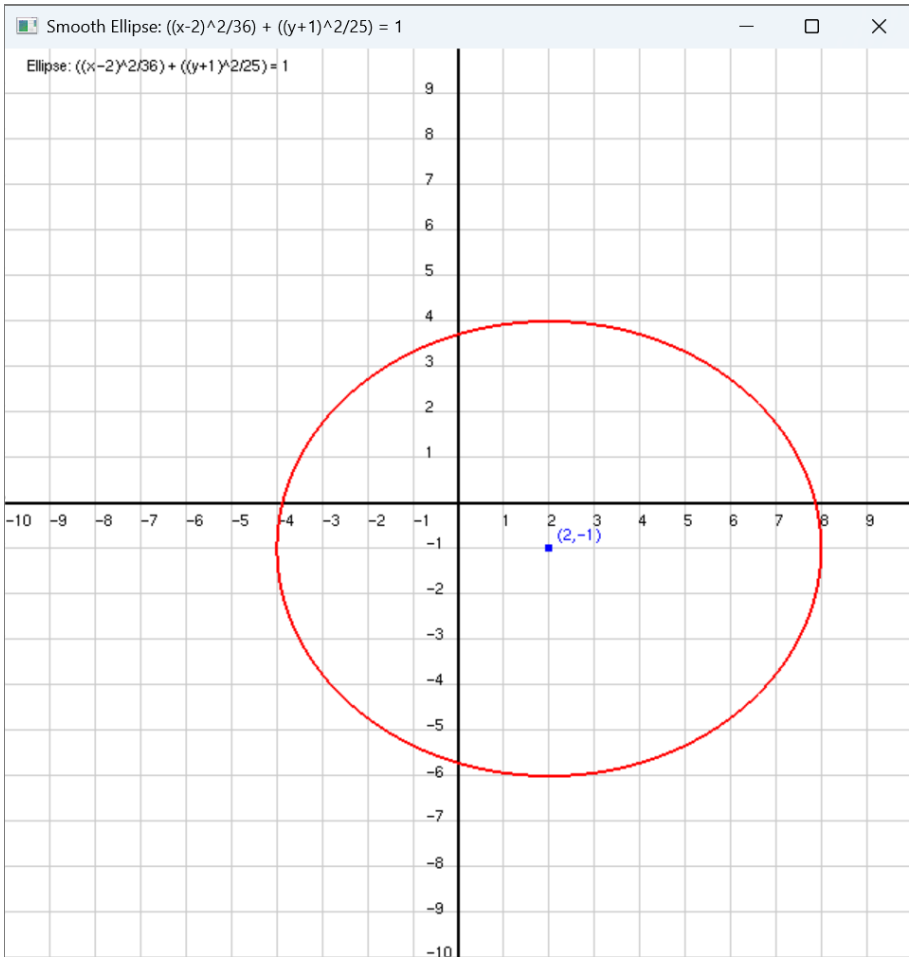
By $\frac{(x-2)^2}{36} + \frac{(y+1)^2}{25} = 1$

This represents an ellipse with:

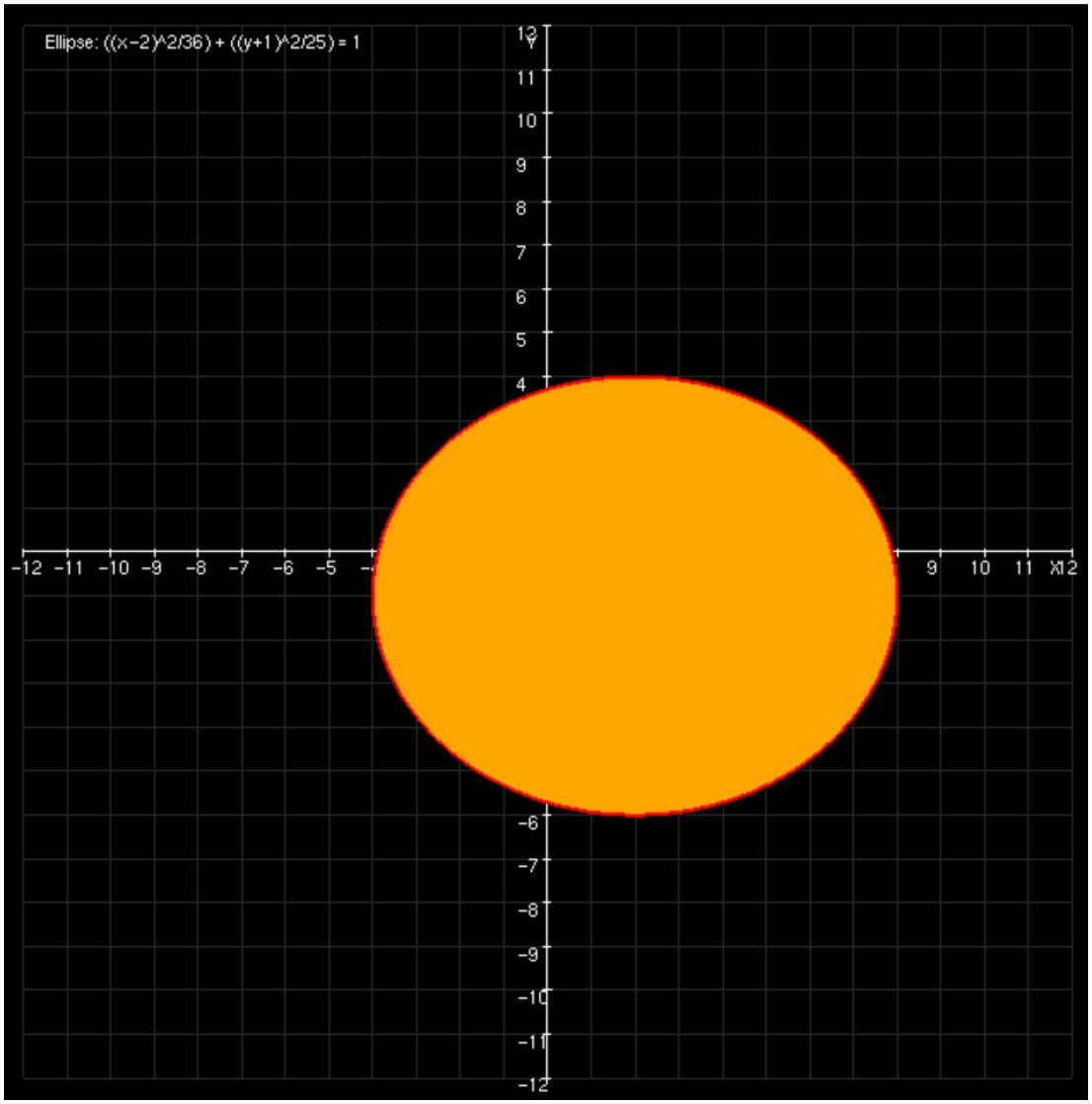
- Center at $(2, -1)$
- Semi-major axis $a = 6$ (along the x-axis)
- Semi-minor axis $b = 5$ (along the y-axis)

The Midpoint Ellipse Algorithm is used to rasterize an ellipse. It works by:

1. Dividing the ellipse into two regions based on the slope of the tangent.
2. Using symmetry to plot points in all four quadrants.
3. Calculating the decision parameter to determine the next pixel.



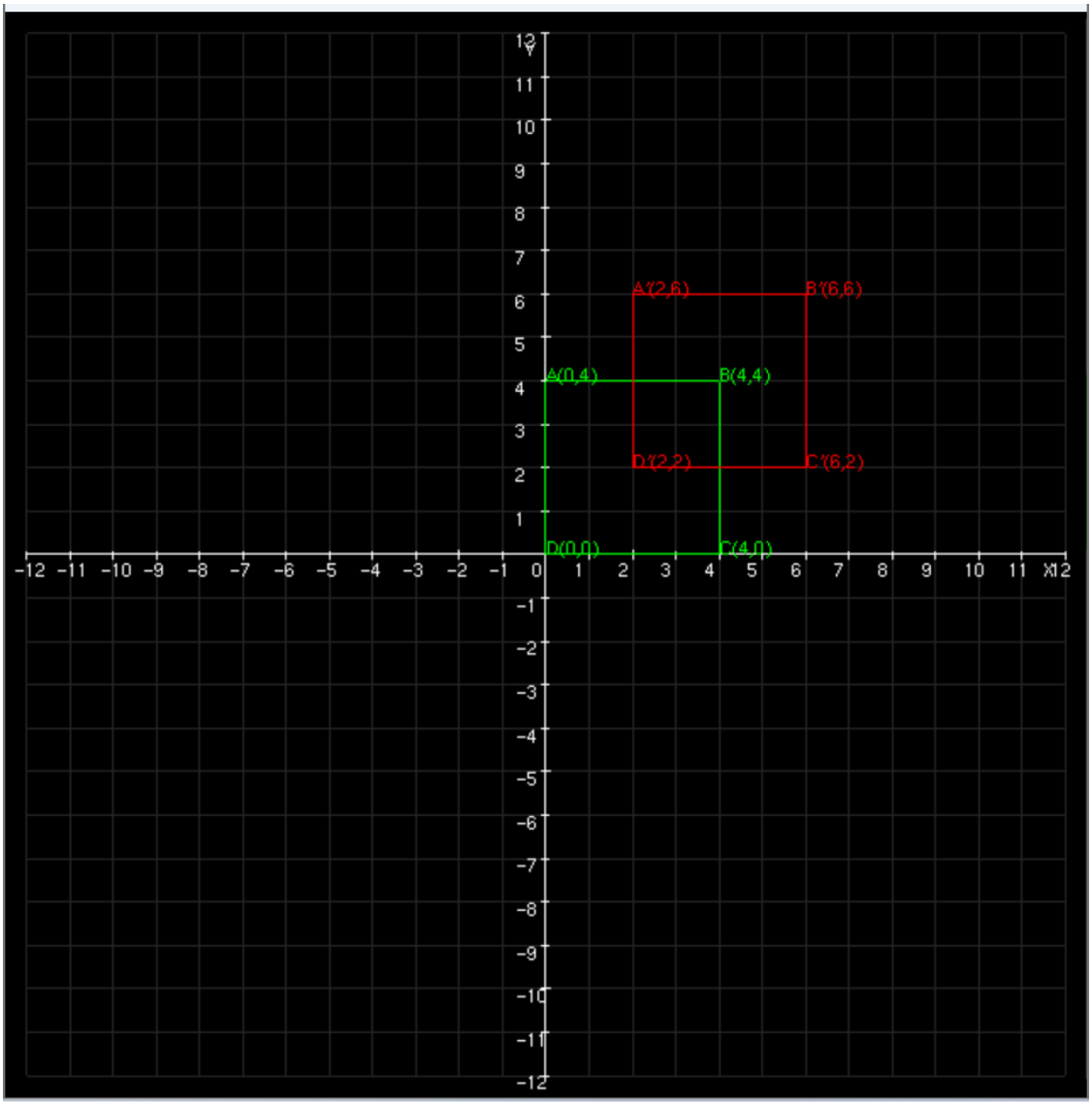
i) Apply the flood-fill algorithm in OpenGL to fill the interior of the above ellipse with orange color (#FFA500)



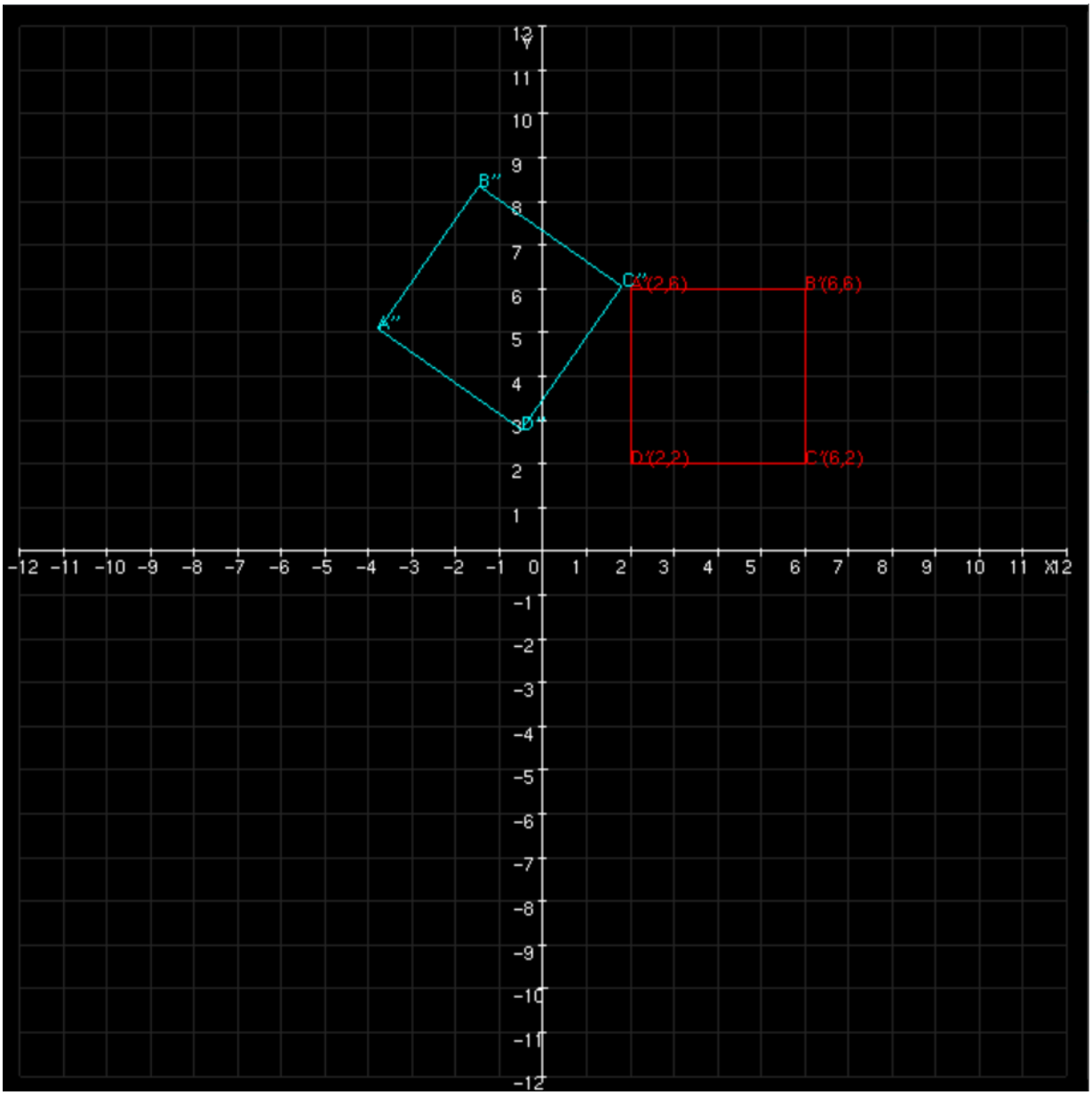
QUESTION 3- SQUARE drawing

a) Using OpenGL Draw a square with coordinate points A (0, 4), B(3, 4), C(4, 0), D(0, 0).

   (i) Apply the translation with distance 2 towards X axis and 2 towards Y axis. Obtain the new coordinates of the square.



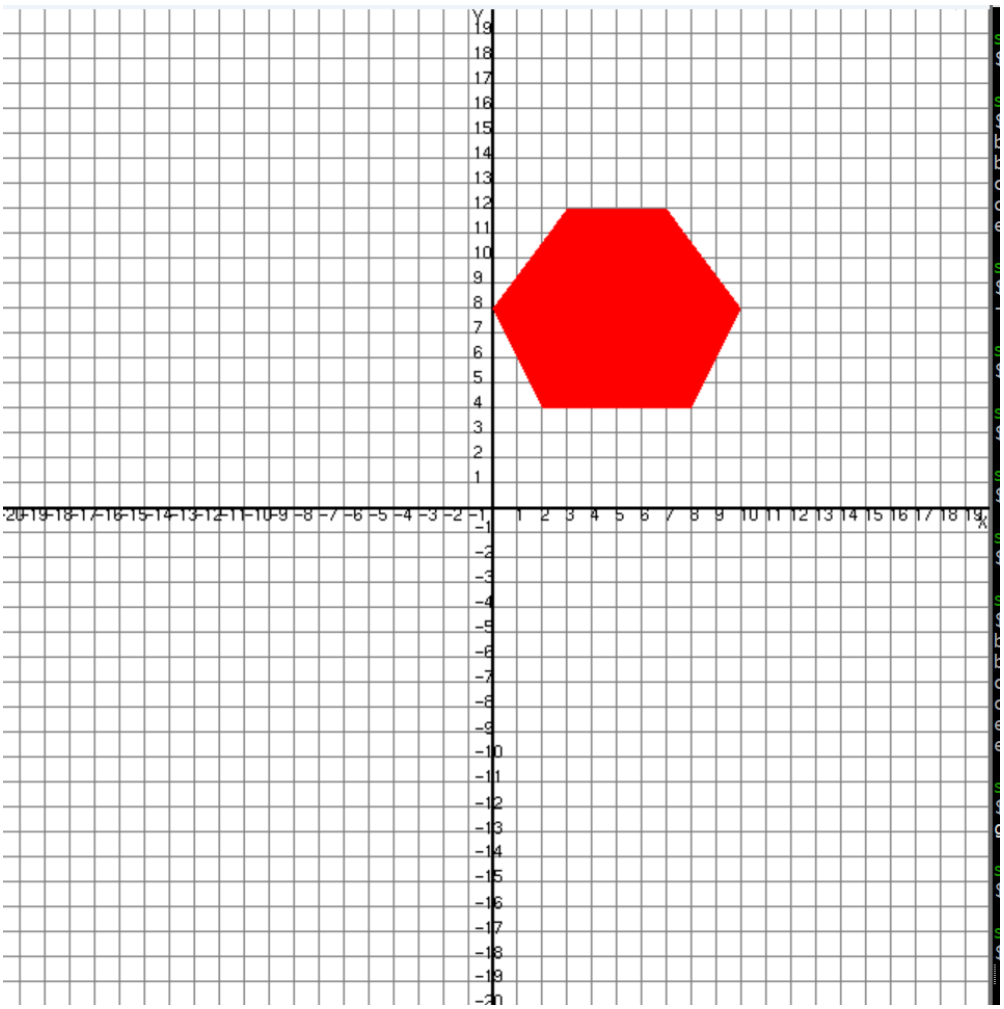
(ii) write an OpenGL program to rotate the translated figure with rotation angle = θ = 55º



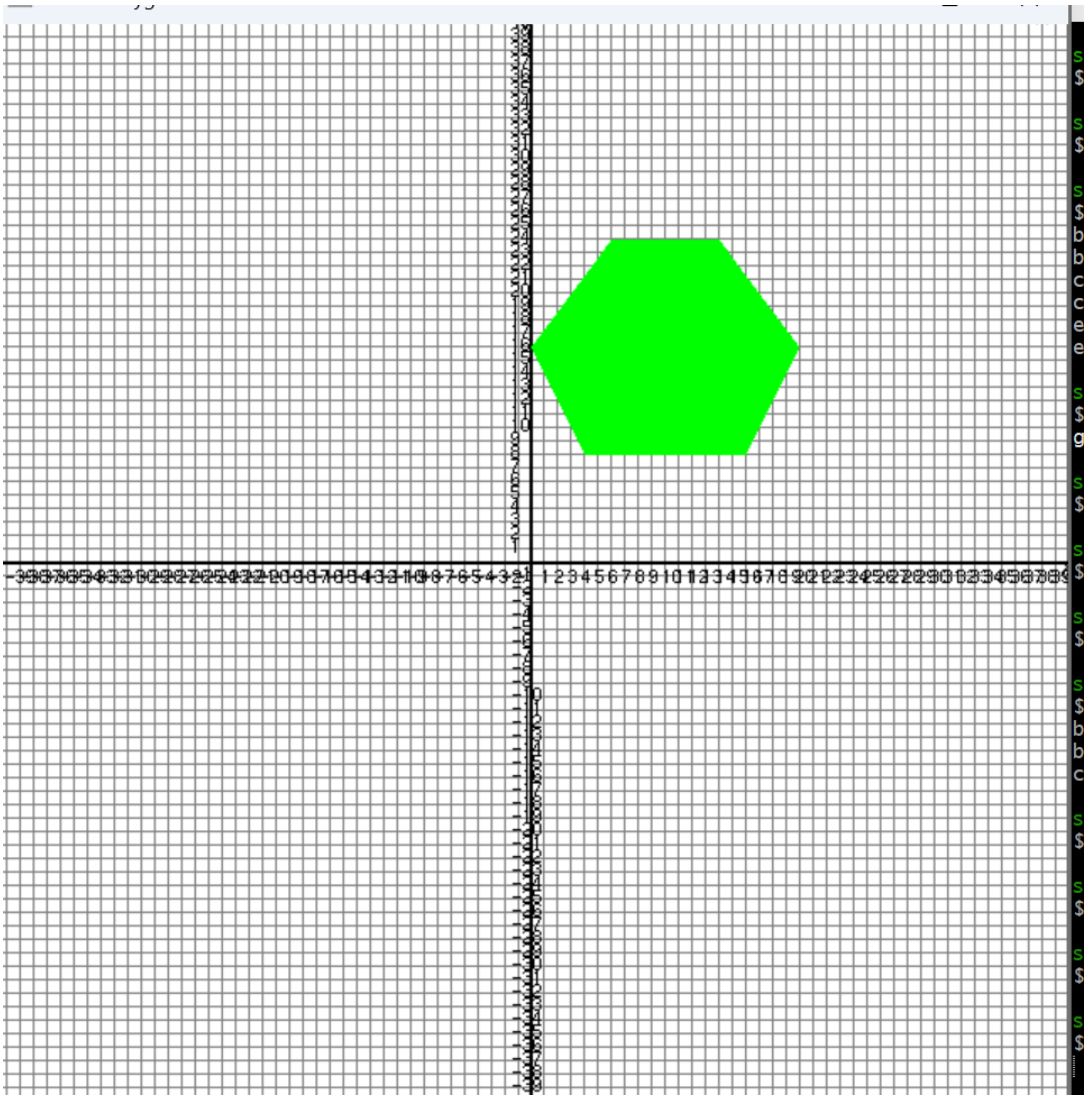
Question 4 - Polygon drawing

a) Using OpenGL draw a filled polygon with the following dimensions (8,4;2,4;0,8;3,12;7,12;10,8) hint (GL POLYGON function) might be useful

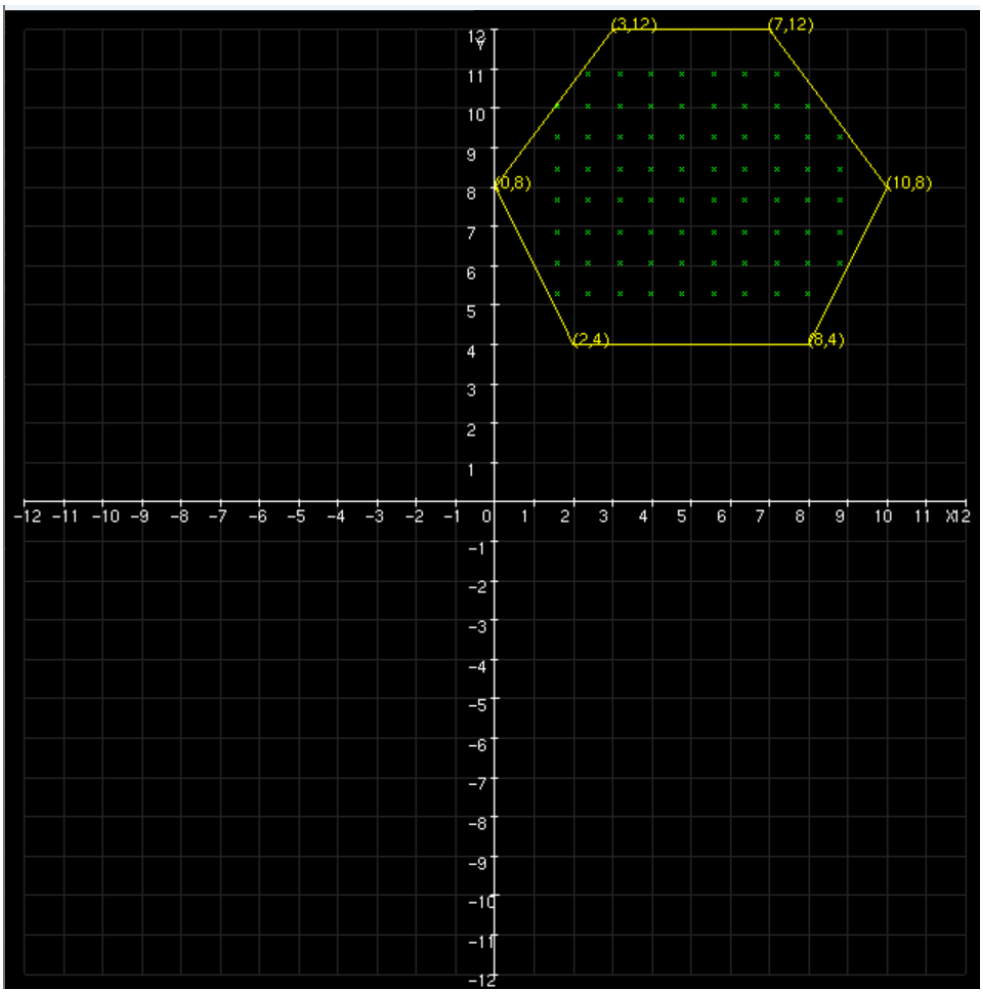
(i). Write a function in OpenGL to fill the polygon above in Red color (#FF0000.)



(ii). Write OpenGL program to scale up (scaling) the polygon by a factor of 2

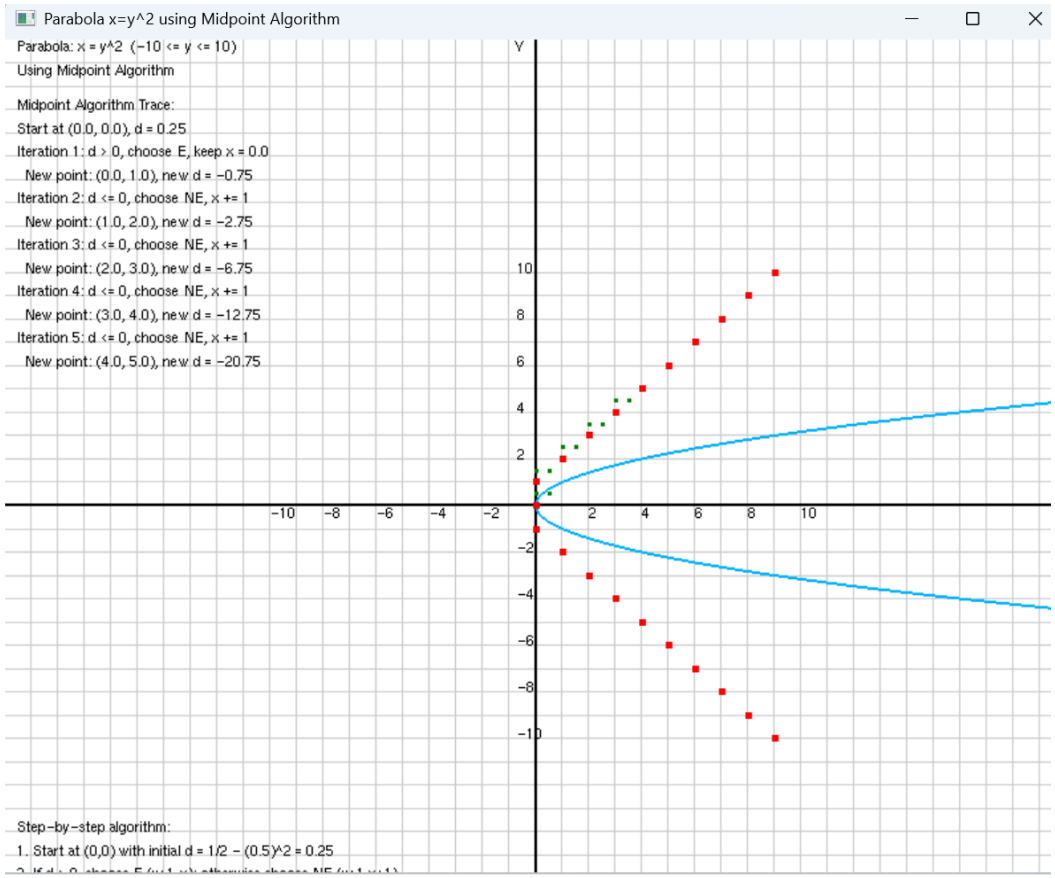


(iii) Write a procedure in OpenGL to fill the interior of the given polygon above with shades of green asterisks



QUESTION 5-PARABOLA Drawing

a) Use the midpoint method and symmetry considerations to scan convert the parabola x=y2 for the interval - 10 ≤ y ≤10 Show the working of the method and implement it using OpenGL



Points generated by the midpoint algorithm:

(0, 0)
(0, 1)
(1, 2)
(2, 3)
(3, 4)
(4, 5)
(5, 6)
(6, 7)
(7, 8)
(8, 9)
(9, 10)